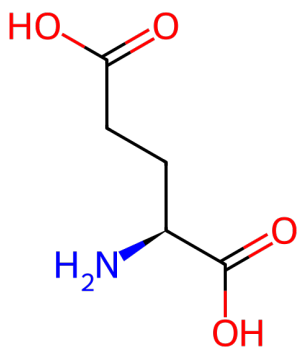
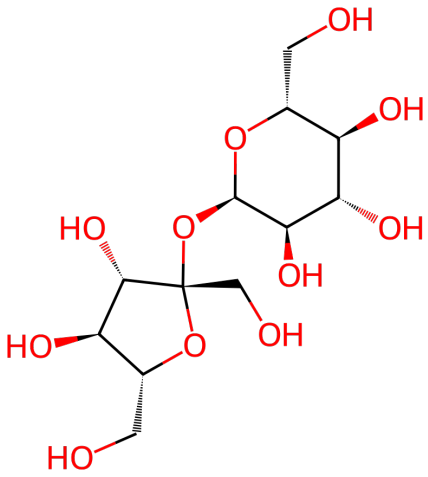
**MOLECULE**




$C_5H_9NO_4$

MolNeuro Retreat 2026

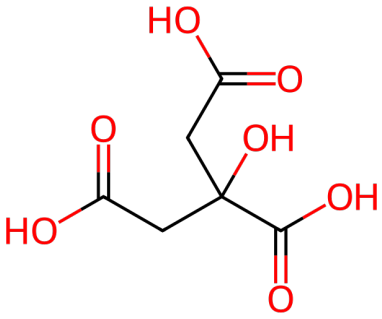
**MOLECULE**




$C_{12}H_{22}O_{11}$

MolNeuro Retreat 2026

**MOLECULE**




$C_6H_8O_7$

MolNeuro Retreat 2026

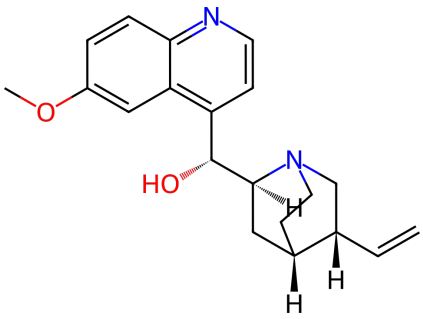
**MOLECULE**

$Na^+$ $Cl^-$

NaCl

MolNeuro Retreat 2026

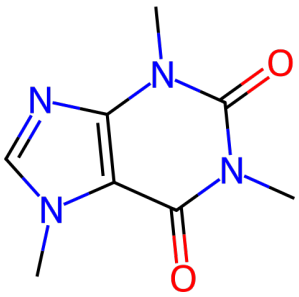
**MOLECULE**




$C_{20}H_{24}N_2O_2$

MolNeuro Retreat 2026

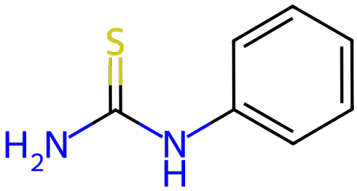
**MOLECULE**




$C_8H_{10}N_4O_2$

MolNeuro Retreat 2026

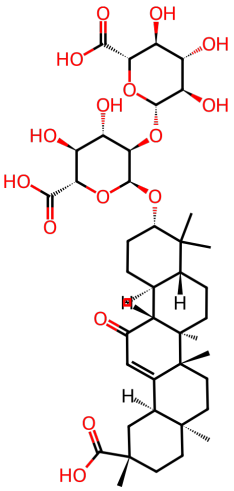
**MOLECULE**




$C_7H_8N_2S$

MolNeuro Retreat 2026

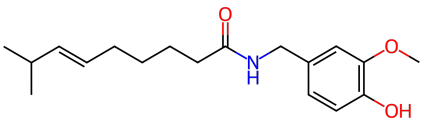
**MOLECULE**




$C_{42}H_{62}O_{16}$

MolNeuro Retreat 2026

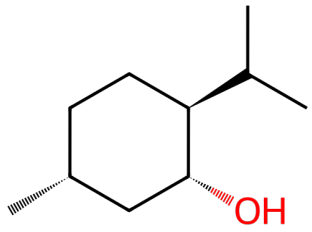
**MOLECULE**




$C_{18}H_{27}NO_3$

MolNeuro Retreat 2026

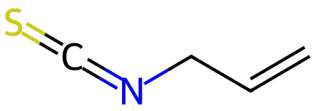
**MOLECULE**




$C_{10}H_{20}O$

MolNeuro Retreat 2026

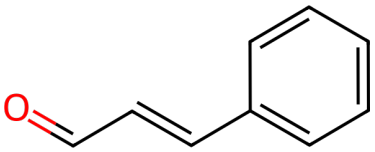
**MOLECULE**




$C_4H_5NS$

MolNeuro Retreat 2026

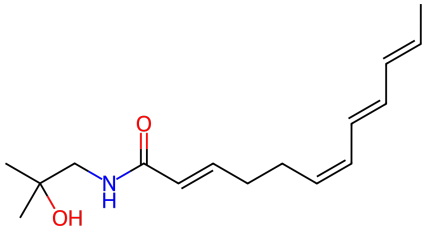
**MOLECULE**




$C_9H_8O$

MolNeuro Retreat 2026

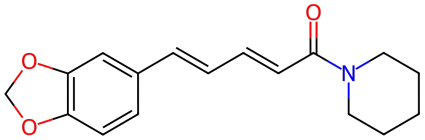
**MOLECULE**




$C_{16}H_{25}NO_2$

MolNeuro Retreat 2026

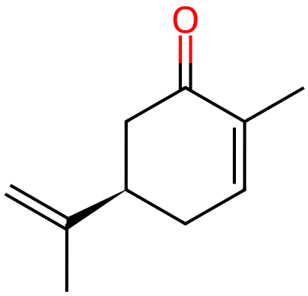
**MOLECULE**




$C_{17}H_{19}NO_3$

MolNeuro Retreat 2026

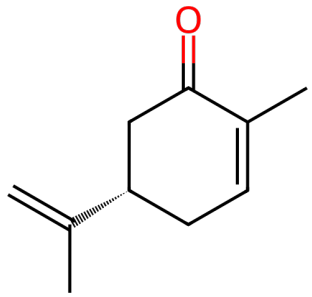
**MOLECULE**




$C_{10}H_{14}O$

MolNeuro Retreat 2026

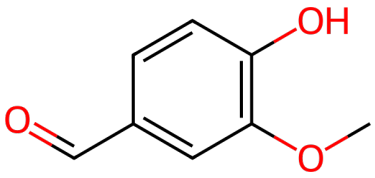
**MOLECULE**




$C_{10}H_{14}O$

MolNeuro Retreat 2026

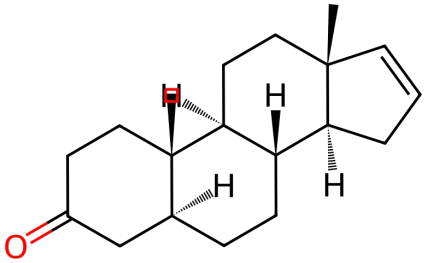
**MOLECULE**




$C_8H_8O_3$

MolNeuro Retreat 2026

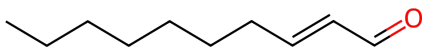
**MOLECULE**




$C_{19}H_{28}O$

MolNeuro Retreat 2026

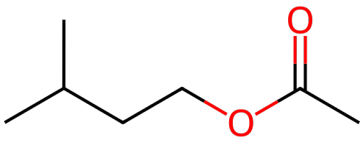
**MOLECULE**




$C_{10}H_{18}O$

MolNeuro Retreat 2026

**MOLECULE**




$C_7H_{14}O_2$

MolNeuro Retreat 2026

FLAVOUR



## Umami

Parmesan · tomato · kombu

RECEPTOR

TAS1R1 / TAS1R3

MolNeuro Retreat 2026

FLAVOUR



## Sweet

Sugar cane · sugar beet · sweets

RECEPTOR

TAS1R2 / TAS1R3

MolNeuro Retreat 2026

FLAVOUR



## Sour

Lemon · lime

RECEPTOR

OTOP1 (proton channel)

MolNeuro Retreat 2026

FLAVOUR



## Salty

Sea salt · crisps

RECEPTOR

ENaC

MolNeuro Retreat 2026

FLAVOUR



## Bitter

Tonic water

RECEPTOR

Several TAS2Rs

MolNeuro Retreat 2026

FLAVOUR



## Bitter

Coffee · tea

RECEPTOR

TAS2R7/10/14/43/46

MolNeuro Retreat 2026

FLAVOUR



## Bitter... or nothing?

Broccoli · Brussels sprouts

RECEPTOR

TAS2R38

MolNeuro Retreat 2026

FLAVOUR



## Intensely sweet

Liquorice root · salmiak

RECEPTOR

TAS1R2 / TAS1R3

MolNeuro Retreat 2026

FLAVOUR



## Burning heat

Chilli peppers

RECEPTOR

TRPV1

MolNeuro Retreat 2026

FLAVOUR



## Cooling

Peppermint · cough drops

RECEPTOR

TRPM8

MolNeuro Retreat 2026

FLAVOUR



## Sharp, up-the-nose

Wasabi · mustard · horseradish

RECEPTOR

TRPA1

MolNeuro Retreat 2026

FLAVOUR



## Warm & spicy

Cinnamon · kanelbullar

RECEPTOR

TRPA1 (+ olfactory receptors)

MolNeuro Retreat 2026

FLAVOUR



## Tingling & numbing

Sichuan pepper

RECEPTOR

KCNK3/9/18 (two-pore $K^+$ channels)

MolNeuro Retreat 2026

FLAVOUR



## Peppery bite

Black pepper · cacio e pepe

RECEPTOR

TRPV1 / TRPA1

MolNeuro Retreat 2026

FLAVOUR



## Spearmint

Spearmint · chewing gum

RECEPTOR

Olfactory receptors

MolNeuro Retreat 2026

FLAVOUR



## Caraway & dill

Dill · caraway · aquavit

RECEPTOR

Olfactory receptors

MolNeuro Retreat 2026

FLAVOUR



## Vanilla

Vanilla pods · ice cream

RECEPTOR

OR10G4

MolNeuro Retreat 2026

FLAVOUR



## Urine? Sandalwood? Nothing?

Boar meat ('boar taint')

RECEPTOR

OR7D4

MolNeuro Retreat 2026

FLAVOUR



## Fresh herb… or soap?

Coriander leaves

RECEPTOR

OR6A2 locus

MolNeuro Retreat 2026

FLAVOUR



## Banana / pear drops

Bananas · pear-drop sweets

RECEPTOR

Olfactory receptors

MolNeuro Retreat 2026